Human-Factor-Coupled Collision Avoidance and Search-and-Rescue in a Multi-Agent Maritime Safety Simulation: A Baltic–North Sea Case Study

Michał Pytel[1], *Paweł Pozorski*[1]

[1] Warsaw University of Technology, Poland

{michal.pytel2, pawel.pozorski}@pw.edu.pl

Abstract

Between 70–96 % of maritime casualties involve human error [1], and heavy weather makes coordination harder by degrading both perception and bridge-to-bridge radio. We present a tick-based agent-based model of Baltic and North Sea traffic. Vessels follow depth- and lane-constrained routes built from open bathymetry and Automatic Identification System (AIS) density data. Encounters are resolved with a Closest Point of Approach (CPA) check and a COLREGs-style give-way rule (International Regulations for Preventing Collisions at Sea). The main modelling choice is that every collision warning succeeds only with probability equal to the product of a local weather factor and a crew-fatigue factor. Collision damage uses DTU SIMCOL [2]; route-network collision frequency is checked with IWRAP Mk II [3]; rescue follows IAMSAR practice [4] with MASSIM/Koopman random search [5, 6] and Ashrafi seasonal degradation [7]. We compare a classical baseline, a weather-aware baseline, and a fatigue-aware Proposed System across four scenarios (30 seeds each; 360 runs) with Mann–Whitney U tests, Holm–Bonferroni correction, and Cliff's δ. Pre-registered fatal-rate and survival thresholds (H1–H3) were not confirmed after correction. Preparedness P_{prep} rose with large confirmed effects in every scenario ($|\delta| \geq 0.90$). Mean Storm Corridor collisions fell by about 28 % versus Baseline A (δ=0.32, not significant after Holm).

Index Terms: maritime safety, multi-agent systems, agent-based modelling, collision avoidance, human factors, search and rescue

1. Introduction

Maritime transport moves most of world trade, but safety remains limited by human factors. The International Maritime Organization (IMO)[1] attributes 70–96 % of incidents to human causes [1]. The European Maritime Safety Agency (EMSA) [8] reports weather as a frequent compounding factor in serious casualties, especially in congested coastal waters. Hull and machinery claims remain large [9].

The International Regulations for Preventing Collisions at Sea (COLREGs) [10][2] assume that crews see the encounter and can agree who gives way. That assumption breaks down under two effects that many maritime ABMs treat only loosely [11, 12, 13]:

- **Weather and radio.** Sea clutter and storms reduce the chance that a marine VHF[3] warning is heard and acted on. Many models assume that once risk is detected, the manoeuvre always happens.
- **Crew fatigue.** Fatigue builds over a watch and follows a circadian pattern [14]. Prior work often keeps fatigue in a side model, without a direct link to whether a warning is executed.

This paper couples those effects in one product: warning success depends on both local weather and crew fatigue. The model runs on real Baltic–North Sea geography (Section 3). Routes are synthesised from bathymetry and Automatic Identification System (AIS)[4] traffic density (Section 3.3). Avoidance uses Closest Point of Approach / Time to CPA (CPA/TCPA)[5] and a simple give-way rule (Section 4). Damage uses the DTU ship-collision model SIMCOL [2] (Section 5); frequency is checked with IALA IWRAP [3] (Section 7); survivors enter a search-and-rescue (SAR)[6] chain (Section 6). The engine is Rust on krABMaga [15]. Figure 1 summarises the causal chain.

2. Objectives and Hypotheses

We test whether linking fatigue management to warning reliability reduces fatal outcomes more than classical, weather-unaware practice. We implement three configurations:

Baseline A (Classical) COLREGs give-way, no weather-aware slowdown, unmanaged fatigue. Storm radio loss still applies inside the storm zone — it is environmental, not a privilege of other methods.

Baseline B (Shore Rest Alerts) Weather-aware speed reduction and slower fatigue build-up; no further mitigation.

Proposed System Smart watch scheduling (lowest fatigue build-rate) on top of Baseline B. Six KPIs are defined in Table 3.

We use them for the following hypotheses:

H1. Proposed reduces the fatal-event rate (Table 3, KPI 1) by $\geq$20 % versus Baseline A ($p < 0.05$, Mann–Whitney U, Cliff's $|\delta| \geq 0.2$).

H2. Proposed improves post-incident survival (KPI 4) by $\geq$15 % versus Baseline A.

H3. Under Storm Corridor, Proposed beats Baseline B on the fatal-event rate ($p < 0.05$).

[1] https://en.wikipedia.org/wiki/International_Maritime_Organization

[2] https://en.wikipedia.org/wiki/International_Regulations_for_Preventing_Collisions_at_Sea

[3] https://en.wikipedia.org/wiki/Marine_VHF_radio

[4] https://en.wikipedia.org/wiki/Automatic_identification_system

[5] Standard radar/AIS encounter metrics: distance and time to the predicted closest point of approach. Overview: https://en.wikipedia.org/wiki/Collision_avoidance

[6] https://en.wikipedia.org/wiki/Search_and_rescue

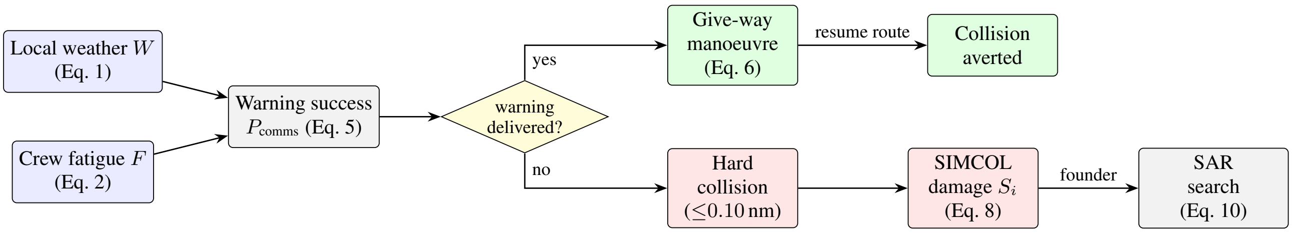

Figure 1: *Causal pipeline. Weather W and fatigue F set warning success P_{comms} (Eq. 5). A delivered warning yields a give-way manoeuvre; a missed warning leaves the encounter to geometry. Hard collisions are scored with SIMCOL; a foundering ship enters SAR.*

3. Simulation Model

This section defines the world, agents, routes, and weather field used by all methods.

3.1. Domain and Time

The domain is the Baltic and North Sea: latitude $\phi \in [50.5°, 66.0°]$ N and longitude $\lambda \in [-5.0°, 31.0°]$ E. An equirectangular[7] map with a mid-latitude correction sets one field unit to one nautical mile[8] at $\phi_0 = 58.25°$:

$$x = (\lambda - \lambda_{\min})\, k_{\text{lon}}, \qquad y = (\phi - \phi_{\min})\, k_{\text{lat}},$$

with $k_{\text{lat}} = 60\,\text{nm}/°$ and $k_{\text{lon}} = 60 \cos\phi_0$. East–west distortion at the box edges is about 6–10 % and is left uncorrected; distances inside the engine are Euclidean nautical miles. Coastline polygons from Natural Earth[9] sit in an R-tree[10] for land and grounding tests. Bathymetry is used only offline for route planning (Section 3.3).

Time steps are $\Delta t = 15$ min (0.25 h). The default run length is $T = 2880$ ticks (30 days). A wall-clock variable starts at 06:00 UTC and drives the circadian fatigue term (Section 4). Each run uses one seeded `SmallRng`, so the same seed yields identical KPI logs.

3.2. Vessels and Voyage Logic

Agents are ships; after an incident, rescue assets appear (Section 6). Ship i has position $\mathbf{p}_i$, waypoint route R_i, travel direction $d_i \in \{+1, -1\}$, heading ψ_i, cruise speed v_i, state $\sigma_i \in \{\text{ACTIVE}, \text{DOCKED}\}$, crew $n_i = 10$, and fatigue $F_i \in [0, 1]$.

Routes are synthesised once (Section 3.3) and stored as `data/ais_paths.json`. Ports are polygons in `data/ports.json`. Entering a port docks the ship. Cruise speed is drawn at spawn ($u \sim \mathcal{U}(0, 1)$):

$$v_i = \begin{cases} 15 + 8u & \text{passenger / ferry / ro-ro / ro-pax} \\ 10 + 4u & \text{tanker} \\ 10 + 6u & \text{cargo / container / bulk} \\ 10 + 10u & \text{otherwise.} \end{cases}$$

Waypoint pursuit. The active target is an avoidance waypoint if one is queued, otherwise the current route waypoint. A waypoint is reached within 0.5 nm. The step length is $s = \min(v_i\, \Delta t,\ \rho)$. Moves use up to five geometrically halved lengths so thin land is not tunnelled; a post-move check reverts grounding to the last valid position. At a port or route end the ship dwells $D \sim \mathcal{U}\{8, 48\}$ ticks, then reverses. Fleet size N is held fixed: when a ship is removed, a new one is spawned on a random synthesised track.

Same-destination following. Co-directional ships with endpoints within 5 nm share a destination. A trailer within 3 nm of a leader matches the leader's speed; within 0.5 nm it holds station. This avoids side-by-side overlap that the give-way rule does not separate.

3.3. Route Generation

Traffic geometry is produced offline so experiments stay reproducible. Depth comes from the EMODnet DTM [16]. Lane attractiveness comes from EMODnet vessel-density maps [17, 12]. Both are fetched once via the OGC Web Coverage Service (WCS)[11] and stored as grids. Ports are a gazetteer of 42 harbours snapped to navigable water.

Each class c has draft T_c and under-keel clearance (UKC$_c$)[12] [18]. A cell of depth h is navigable iff $h \geq T_c + \text{UKC}_c + m_s$ with buffer $m_s = 2$ m (Table 1). Deep-draft very large crude carriers (VLCCs)[13] are excluded (Danish Straits).

Table 1: *Minimum transit depth by vessel class (draft + underkeel clearance).*

Class	Draft T_c (m)	UKC$_c$ (m)	Required h (m)
Passenger	6.5	1.5	8
Cargo	11.0	2.0	13
Tanker	14.0	3.0	17

Between sampled port pairs, a weighted A* planner [19] runs on a 1 nm grid. Cell cost uses

$$\mu = 1 + w_h\, s_h + w_d\, s_d + w_\ell\, (1 - a_\ell),$$

penalising shallow margins (s_h), proximity to shore/shoal (s_d), and off-lane cells (a_ℓ). Default weights are $(w_h, w_d, w_\ell) = (3, 4, 2.5)$. Paths are simplified under a depth constraint and checked against raw bathymetry every 0.15 nm. Seeded cost jitter keeps outputs deterministic per seed. The default network has 80 routes (35 cargo, 25 passenger, 20 tanker).

[7] `https://en.wikipedia.org/wiki/Equirectangular_projection`
[8] `https://en.wikipedia.org/wiki/Nautical_mile`
[9] `https://en.wikipedia.org/wiki/Natural_Earth`
[10] `https://en.wikipedia.org/wiki/R-tree`
[11] `https://en.wikipedia.org/wiki/Web_Coverage_Service`
[12] `https://en.wikipedia.org/wiki/Under_keel_clearance`
[13] `https://en.wikipedia.org/wiki/Very_large_crude_carrier`

3.4. Weather Field

A 50×50 grid holds sea state s, visibility v, and wind u in $[0, 1]$. They fuse into hazard $W_{ij} \in [0, 1]$:

$$W = \mathrm{clip}\left(\frac{w_s\, s + w_v\,(1 - v) + w_u\, u}{w_s + w_v + w_u},\ 0,\ 1\right), \tag{1}$$

with $(w_s, w_v, w_u) = (1.0, 0.5, 0.8)$. Each ship samples the nearest cell. The field advances each tick through a calm/storm regime chain, Lagrangian storm cells, AR(1)[14] background fields, overlays, coupling, and smoothing (parameters in Appendix A).

Independently, Storm Corridor places a moving storm of radius r_{storm} that drifts at 0.30 nm/tick along a fixed track (English Channel → North Sea → Skagerrak → Kattegat → Baltic → Gdańsk). Inside the zone, communication and (for non-A methods) speed are degraded.

4. Crew Fatigue and Collision Avoidance

Fatigue and avoidance form the mechanism behind H1–H3: warnings fail when weather is bad and crews are tired.

4.1. Fatigue Dynamics

Fatigue $F_i \in [0, 1]$ updates every tick. The circadian drive peaks near 03:00:

$$c(t_{\text{utc}}) = \tfrac{1}{2}\left(1 - \cos\left(\tfrac{2\pi(t_{\text{utc}} - 3)}{24}\right)\right) \in [0, 1].$$

While ACTIVE, with local hazard W,

$$\Delta F = \left(\beta_{\text{base}} + \beta_W\, W + \beta_c\, c(t_{\text{utc}})\right) m(\text{method}), \tag{2}$$

with $(\beta_{\text{base}}, \beta_W, \beta_c) = (0.0015,\ 0.0040,\ 0.0008)$ and

$$m(\text{method}) = \begin{cases} 1.00 & \text{Baseline A} \\ 0.82 & \text{Baseline B} \\ 0.65 & \text{Proposed,} \end{cases} \tag{3}$$

then $F_i \leftarrow \min(1,\ F_i + \Delta F)$. While DOCKED, recovery is 0.025 per tick.

Above threshold $F^* = 0.40$, fatigue reduces warning reliability with maximum penalty $\kappa = 0.65$:

$$g(F_i) = \begin{cases} 1, & F_i \le F^* \\ 1 - \kappa\, \dfrac{F_i - F^*}{1 - F^*}, & F_i > F^* \end{cases} \in [0.35,\ 1]. \tag{4}$$

4.2. CPA Detection and Give-Way

Each ACTIVE pair is screened with linear CPA. With relative position $\Delta\mathbf{p}$ and relative velocity $\Delta\mathbf{v}$,

$$t^* = -\frac{\Delta\mathbf{p} \cdot \Delta\mathbf{v}}{\|\Delta\mathbf{v}\|^2}, \qquad d_{\text{CPA}} = \left\|\Delta\mathbf{p} + t^*\, \Delta\mathbf{v}\right\|,$$

where t^* is in ticks. A manoeuvre is considered only if $\|\Delta\mathbf{p}\| \le 20$ nm, $d_{\text{CPA}} \le 5$ nm, $0 < t^* \le 12$ ticks, and the give-way ship is not in cooldown.

[14] https://en.wikipedia.org/wiki/Autoregressive_model

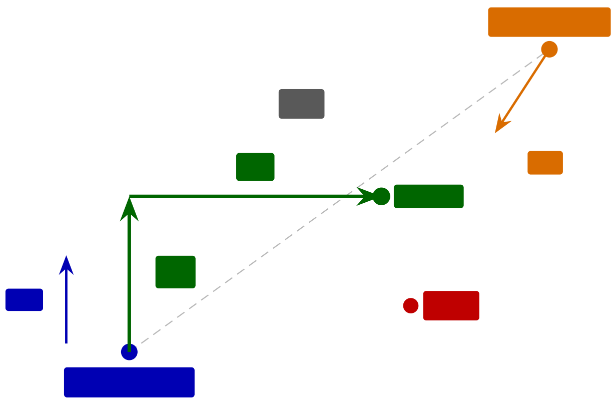

Figure 2: *Give-way geometry. Ship a steers to a temporary waypoint $\mathbf{w}_{avoid}$ placed L_f ahead and L_o to starboard (Eq. 6). The manoeuvre runs only if the warning is delivered (Eq. 5).*

The lower-id ship gives way (turn to starboard); the other stands on. This is a reproducible surrogate for COLREGs stand-on/give-way assignment [10, 11]. The manoeuvre runs only if the warning is delivered:

$$P_{\text{comms}} = \underbrace{\min\left(q(\mathbf{p}_a),\ q(\mathbf{p}_b)\right)}_{\text{weather / storm}} \cdot \underbrace{\min\left(g(F_a),\ g(F_b)\right)}_{\text{fatigue (Eq. 4)}}, \tag{5}$$

where $q(\mathbf{p})$ is nominal link reliability, reduced to the storm-comms rate (0.08 inside Storm Corridor) when $\mathbf{p}$ is in the storm zone. A uniform draw below P_{comms} executes the manoeuvre; otherwise geometry alone decides the encounter. Equation 5 is the central coupling of the model.

A give-way ship receives one temporary waypoint $L_f = 3$ nm ahead and $L_o = 5$ nm to starboard of heading ψ (Fig. 2):

$$\mathbf{w}_{\text{avoid}} = \mathbf{p} + L_f\,(\sin\psi,\ \cos\psi) + L_o\,(\cos\psi,\ -\sin\psi). \tag{6}$$

A short message log records the warning exchange. A 3-tick cooldown prevents thrashing.

4.3. Optional Force-Field Track

As an optional, default-off switch, the discrete waypoint of Eq. 6 can be replaced by a continuous repulsion field in the style of Xiao et al. [20]:

$$\mathbf{F}_{\text{rep}} = \sum_{k:\, d_k < d_o} \frac{k_{\text{rep}}}{d_k^p}\, \hat{\mathbf{n}}_k,$$

with exponent $p = 2$ by default. Source effect distances are micro-scale relative to a 15-minute basin tick, so magnitudes are rescaled while keeping the reported head-on to overtaking ratio. All definitive experiments keep this track off.

5. Collision Consequence (SIMCOL)

A **hard collision** is logged when two ACTIVE ships close within 0.10 nm (≈185 m), once per encounter (rising edge). Contacts within 1 nm of a port are ignored as harbour manoeuvring under vessel traffic services (VTS)[15]. Damage uses DTU SIMCOL (ship collision damage / survivability) [2].

[15] https://en.wikipedia.org/wiki/Vessel_traffic_service

Absorbed energy follows Pedersen and Zhang [21, 2] on the normal closing component:

$$E_{\text{abs}} = \tfrac{1}{2}\,\frac{M'_a\,M'_b}{M'_a + M'_b}\,V_n^2, \qquad V_n = \big|(\mathbf{v}_b - \mathbf{v}_a)\cdot\hat{\mathbf{n}}\big|, \tag{7}$$

with added-mass factors $m_{\text{sway}} = 0.85$ and $m_{\text{surge}} = 0.05$ [2]. Minorsky's correlation [22] links energy to damaged volume. Relative penetration t/B drives a survival factor shaped like SOLAS Chapter II-1 probabilistic damage stability [23][16]:

$$S_i = \begin{cases} 1, & t/B \le r_0 \\ \left(\dfrac{r_1 - t/B}{r_1 - r_0}\right)^{0.25}, & r_0 < t/B < r_1 \\ 0, & t/B \ge r_1, \end{cases} \tag{8}$$

with $r_0 = 0.10$, $r_1 = 0.30$. This is an accepted surrogate: the engine does not carry per-ship residual-stability curves. Expected overboard count is

$$\mathbb{E}[\text{overboard}] = n_b\,\big(1 - S_i\big), \tag{9}$$

drawn as independent Bernoulli trials. If $S_i < S^\dagger = 0.50$, the struck ship founders and the whole crew enters the liferaft (EVAC). If it stays afloat, only the overboard draw enters the man-overboard chain (Section 6.3).

Encounter type is diagnostic, from the folded heading angle θ:

$$\text{type} = \begin{cases} \text{side-to-side (overtaking)}, & \theta < 45^\circ \\ \text{front-to-side (crossing)}, & 45^\circ \le \theta < 135^\circ \\ \text{head-on}, & \theta \ge 135^\circ. \end{cases}$$

Near-parallel hits release little energy in Eq. 7; head-on and crossing hits drive deeper penetration.

6. Search and Rescue

Foundering ($S_i < S^\dagger$) starts the liferaft chain, following International Aeronautical and Maritime Search and Rescue (IAMSAR) Manual practice [4][17]. A rescue asset leaves the nearest port after a mobilisation delay of 2 ticks by default: helicopter (80 kn) beyond 40 nm offshore, otherwise patrol (25 kn). A second chain handles persons overboard from afloat ships (Section 6.3). Figure 3 shows the vessel state machine. Grounding only reverts position; it does not force evacuation.

6.1. Drift and Detection (MASSIM)

Detection follows the Koopman random-search model used in the MASSIM maritime-search simulation [5, 6]. With sweep width w, search speed v_s, and area A,

$$\text{CDP} = 1 - \exp\left(-\,\frac{n\,w\,v_s\,\tau}{A}\right) = 1 - e^{-C}. \tag{10}$$

Search area grows as $A(\tau) = \pi\,(r_0 + u_d\,\tau)^2$ with initial radius r_0 and drift u_d. Each on-scene tick draws Bernoulli detection at coverage $1 - e^{-\mathrm{d}C}$.

[16] https://en.wikipedia.org/wiki/International_Convention_for_the_Safety_of_Life_at_Sea

[17] https://en.wikipedia.org/wiki/International_Aeronautical_and_Maritime_Search_and_Rescue_Manual

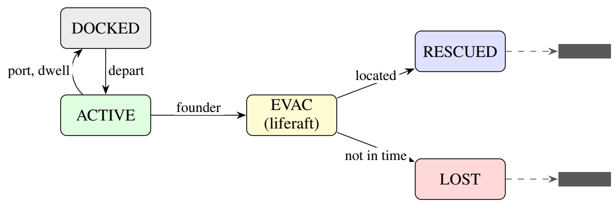

Figure 3: *Vessel state machine. Foundering collision → EVAC; SAR yields RESCUED (Eq. 10) or LOST. Terminal ships respawn to keep fleet size. KPI fatalities count only unrecovered persons-in-water (Section 6.3).*

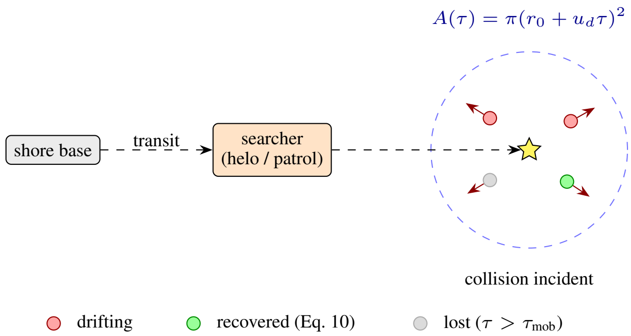

Figure 4: *Man-overboard search. Casualties (Eq. 9) drift separately; one searcher applies Eq. 10 per person. Unrecovered people within τ_{mob} are the fatal KPI events.*

6.2. Seasonal Degradation (Ashrafi)

Asset performance follows Ashrafi et al. [7]. Wind-chill uses

$$\text{wci} = 13.12 + 0.6215\,T - 11.37\,U^{0.16} + 0.3965\,T\,U^{0.16}.$$

Months map to severity groups A–D (May–August mildest; Dec–Feb harshest). Relative transit/search speeds are approximately 1.00/0.95/0.77/0.60 (helicopter) and 1.00/0.80/0.40/0.21 (patrol); boarding times lengthen with severity [7]. Default simulation month is July (group A).

6.3. Man-Overboard Search

Overboard casualties (Eq. 9) form one incident of persons-in-water, each with its own drift bearing (Fig. 4), following Karatas et al. [5]. One searcher homes on the cluster centroid and applies Eq. 10 independently per person. Anyone not recovered within $\tau_{\text{mob}} = 12$ ticks (≈3 h) is lost. **This is the only path that increments the fatal KPI.** Liferaft LOST is logged but does not add to `fatal_per_1k_hrs`. Liferaft endurance is 96 ticks (24 h).

7. External Collision-Frequency Check (IWRAP)

Internal KPIs compare methods. To place traffic intensity on a known scale, the committed route network is analysed with IALA's IWRAP Mk II waterway risk model [3, 24][18]:

$$N_c = N_a\,P_c.$$

[18] https://en.wikipedia.org/wiki/International_Association_of_Marine_Aids_to_Navigation_and_Lighthouse_Authorities

Head-on, overtaking, and crossing candidate counts follow the usual IWRAP forms (flows Q, beams B, speeds V, Pedersen diameter for crossings [3, 25]). Causation probabilities are the IWRAP defaults: $P_c^{\text{head-on}} = 0.50 \times 10^{-4}$, $P_c^{\text{overtaking}} = 1.10 \times 10^{-4}$, $P_c^{\text{crossing}} = 1.30 \times 10^{-4}$.

Each batch row stamps N_c from the static synthesised routes, the scenario fleet size, and a method speed scale (weather-aware methods slow traffic). We do not export per-run AIS tracks. For the present network, N_c is on the order of 0.01 collisions/yr, inside the usual <1/yr/fairway acceptance band.

8. Experimental Design

8.1. Methods and Scenarios

The three methods of Section 2 differ as follows:

- **Baseline A:** no weather-aware slowdown (storm speed factor = 1); unmanaged fatigue ($m = 1.00$).
- **Baseline B:** hazard and storm slowdowns; $m = 0.82$.
- **Proposed:** same slowdowns as B; $m = 0.65$.

SIMCOL parameters are identical across methods. Storm-zone radio loss applies to all methods. Table 2 lists the four scenarios. Each condition uses 30 seeds and 2880 ticks ($4 \times 3 \times 30 = 360$ runs).

8.2. Key Performance Indicators

Table 3 defines the six hypothesis KPIs. Ship-hours add 0.25 h per ACTIVE ship per tick. A *fatality* in the KPIs is an unrecovered person-in-water (Section 6.3).

Crew preparedness uses method awareness $\alpha_M \in \{2.0\,(\text{A}),\ 1.3\,(\text{B}),\ 1.0\,(\text{Proposed})\}$:

$$P_{\text{prep}} = \max(0,\ 1 - \alpha_M W) \cdot \max(0,\ 1 - 0.70\,F).$$

Diagnostic counters (collision-type mix, MOB tallies) are logged but not entered into the Holm battery.

8.3. Statistical Analysis

For each (KPI × scenario × method pair) we use a two-tailed Mann–Whitney U test[19] with Cliff's δ [26]. Multiplicity uses Holm–Bonferroni[20] step-down over the full battery. A result is *significant* if $p_{\text{corr}} < 0.05$ and *confirmed* if also $|\delta| \geq 0.2$. Pairs are (A vs Proposed), (B vs Proposed), and (A vs B). Seeds are matched across conditions.

9. Implementation

The engine is Rust on krABMaga [15]. An Axum HTTP API launches factorial batches (Rayon) and streams progress with Server-Sent Events (SSE)[21]. Each finished run writes a JSONL tick log (every ten ticks), a manifest, and a SQLite[22] results row. A React workbench (deck.gl / MapLibre) replays completed runs.

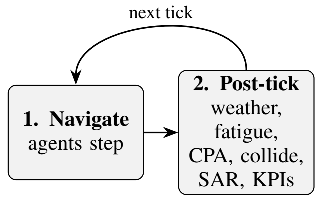

Figure 5: *Macro-tick order. Agents navigate, then the post-tick pipeline updates weather, human factors, encounters, and outcomes.*

9.1. Tick Ordering

Within a tick, vessel agents step first (krABMaga order 200), then a post-tick agent (order 1000) runs the shared physics pipeline shown in Fig. 5: weather advance; grounding revert; hazard and storm speed damping; fatigue; CPA avoidance and comms draws; following; rising-edge collisions and SIMCOL; SAR/MOB; fleet top-up; KPI accumulate; optional JSONL frame. Navigation therefore uses weather and avoidance state from the previous tick's post-tick pass.

9.2. Speed Reduction

Baseline B and Proposed interpolate the just-moved position toward the last valid position by

$$f_W = \max\big(0.30,\ 1 - 0.45\,W\big).$$

Baseline A ignores this factor. Inside the storm zone an extra factor applies: 1.0 for Baseline A, 0.45 in Storm Corridor for the other methods (default non-corridor storm factor 0.70 when a storm is enabled).

10. Results

The definitive batch ran 30 seeds × 4 scenarios × 3 methods (360 runs). Confirmation requires $p_{\text{corr}} < 0.05$ and $|\delta| \geq 0.2$. Figure 6 shows per-run KPI distributions; Figure 7 summarises effect sizes for the pre-registered contrasts.

10.1. Calibration and Storm Contrast

Under Baseline A, Calm Passage mean collisions were 0.60 per 1 000 ship-hours (inside the 0.5–1.5 band). Storm Corridor raised that rate to 1.48 (2.48× calm), restoring a clear weather contrast after the depth+lane route rebase.

10.2. Hypotheses H1–H3

Table 4 summarises the pre-registered contrasts. **H1** and **H2** are not confirmed: fatal rates and survival ratios sit near floor/ceiling in Calm Passage, Storm Corridor, and Blind Shore, so tests have little power. Deep Water Rescue has a higher fatal base rate; Proposed reduces mean fatals from 0.038 (Baseline A) to 0.023 (≈39 %), but survival is almost unchanged and neither contrast survives Holm ($p_{\text{corr}} = 1$, $|\delta| \leq 0.02$). **H3** is non-significant (means 0.0022 vs 0.0024).

10.3. Mechanism and Collisions

Mean P_{prep} is higher for Proposed than for either baseline in every scenario, with confirmed effects ($|\delta| \in [0.90, 1.00]$). Mean Storm Corridor collisions fall from 1.48 to 1.07 versus Base-

[19] https://en.wikipedia.org/wiki/Mann-Whitney_U_test

[20] https://en.wikipedia.org/wiki/Holm-Bonferroni_method

[21] https://en.wikipedia.org/wiki/Server-sent_events

[22] https://en.wikipedia.org/wiki/SQLite

Table 2: *Scenario summary (*30 *seeds,* 2880 *ticks/run).*

Scenario	Key conditions	Ships	Hyp.
Calm Passage	Calm weather; calibration reference (target band $\approx$0.5–1.5 collisions / 1 k ship-hrs under Baseline A).	20	Calib.
Storm Corridor	Moving storm; storm-comms rate 0.08; non-A speed factor 0.45.	25	H1, H3
Blind Shore	Fleet-wide comms 0.55; tighter avoidance berth ($L_o = 3\,\mathrm{nm}$).	25	H1
Deep Water Rescue	Sparse fleet; SAR mobilisation 4 ticks; helo/patrol $55/16\,\mathrm{kn}$.	15	H2

Table 3: *KPI definitions and hypothesis links.*

#	KPI	Definition	Hyp.
1	`fatal_per_1k_hrs`	Fatalities $\div$ ship-hrs $\times 10^3$	H1, H3
2	`collision_per_1k_hrs`	Collisions $\div$ ship-hrs $\times 10^3$	Calib.
3	`evac_activation_rate`	Evac activations $\div$ ship-hrs $\times 10^3$	—
4	`survival_ratio`	$1-$ fatalities / crew exposed in collisions	H2
5	`avg_tta_hours`	Mean rescue time-to-arrival	H2
6	`mean_p_prep`	Time-averaged crew preparedness	mech.

Table 4: *Pre-registered hypothesis outcomes (definitive* 30*-seed batch).*

Hyp.	Contrast	Verdict	Cliff's δ
H1	Fatal rate, Proposed vs A	Not confirmed	$\lvert\delta\rvert \le 0.07$
H2	Survival ratio, Proposed vs A	Not confirmed	$\lvert\delta\rvert \le 0.07$
H3	Fatals under Storm, Proposed vs B	Not confirmed	$\delta \approx 0.00$

line A ($\approx$28 %, δ=0.32), but that contrast is not significant after family-wise Holm correction.

11. Discussion

The core claim is Eq. 5: a warning succeeds only when weather and both crews allow it. Methods differ by fatigue build-rate m (Eq. 3), weather/storm speed awareness, and preparedness weights α_M. The batch confirms the preparedness lever and a directional storm-collision benefit, but not the pre-registered fatal/survival thresholds. Fatal events are rare outside Deep Water Rescue, so H1–H3 are under-powered even when point estimates move in the expected direction.

Accredited layers add external structure: SIMCOL for damage [2], IWRAP for network frequency [3], and MASSIM/Ashrafi for rescue [5, 7]. The survival factor S_i (Eq. 8) is a SOLAS-shaped surrogate, not a full residual-stability calculation.

Main limitations: synthesised routes trade some traffic realism for control; the lower-id give-way rule simplifies COLREGs; force-field avoidance is off by default; KPI fatalities count only MOB timeouts; IWRAP stamps use static routes rather than per-run tracks. Calibration under Baseline A Calm Passage (0.60 collisions per 1 000 ship-hours) sits near published AIS near-miss bands [12, 13]. Longer runs or collision-rate primary endpoints would raise power on fatal contrasts.

12. References

[1] International Maritime Organization, "Human element – addressing the causes of accidents and incidents at sea," International Maritime Organization, London, UK, IMO Circular MSC-MEPC.7/Circ.7, 2018. [Online]. Available: https://wwwcdn.imo.org/localresources/en/OurWork/HumanElement/Documents/MSC-MEPC.7-Circ.7.pdf

[2] M. Lützen, "Ship collision damage," Ph.D. dissertation, Technical University of Denmark, Department of Mechanical Engineering, Kongens Lyngby, Denmark, 2001.

[3] P. Friis-Hansen, "Basic modelling principles for prediction of collision and grounding frequencies (IWRAP Mk II working document)," Technical University of Denmark, Kongens Lyngby, Denmark, Tech. Rep., 2008, rev. 4.

[4] International Maritime Organization, "International aeronautical and maritime search and rescue (IAMSAR) manual," IMO / ICAO, London, UK, Tech. Rep., 2019.

[5] M. Karatas, B. S. Onggo, and M. M. Gunal, "Using simulation to measure the performance of a search operation for a man overboard," in *Proceedings of the 2018 Winter Simulation Conference (WSC)*. Gothenburg, Sweden: IEEE, 2018, pp. 2742–2753.

[6] B. O. Koopman, *Search and Screening: General Principles with Historical Applications*. New York: Pergamon Press, 1980.

[7] B. Ashrafi, G. Kim, M. Naseri, J. Barabady, S. Dhar, G. Heo, and S. Baek, "An agent-based modelling framework for performance assessment of search and rescue operations in the Barents Sea," *Safety in Extreme Environments*, vol. 6, pp. 183–200, 2024.

[8] European Maritime Safety Agency, "Annual overview of marine casualties and incidents 2023," EMSA, Lisbon, Portugal, Tech. Rep., 2023. [Online]. Available: https://www.emsa.europa.eu/accident-investigation-publications/annual-overview.html

[9] International Union of Marine Insurance, "IUMI ocean hull statistics 2023," IUMI, Hamburg, Germany, Tech. Rep., 2023. [Online]. Available: https://iumi.com/statistics

[10] International Maritime Organization, "Convention on the international regulations for preventing collisions at sea, 1972 (COLREGs)," International Maritime Organization, London, UK, Tech. Rep., 1972. [Online]. Available: https://www.imo.org/en/About/Conventions/Pages/COLREG.aspx

[11] C.-k. Tam, R. Bucknall, and A. Greig, "Review of collision avoidance and path planning methods for ships in close range encounters," *Journal of Navigation*, vol. 62, no. 3, pp. 455–476, 2010.

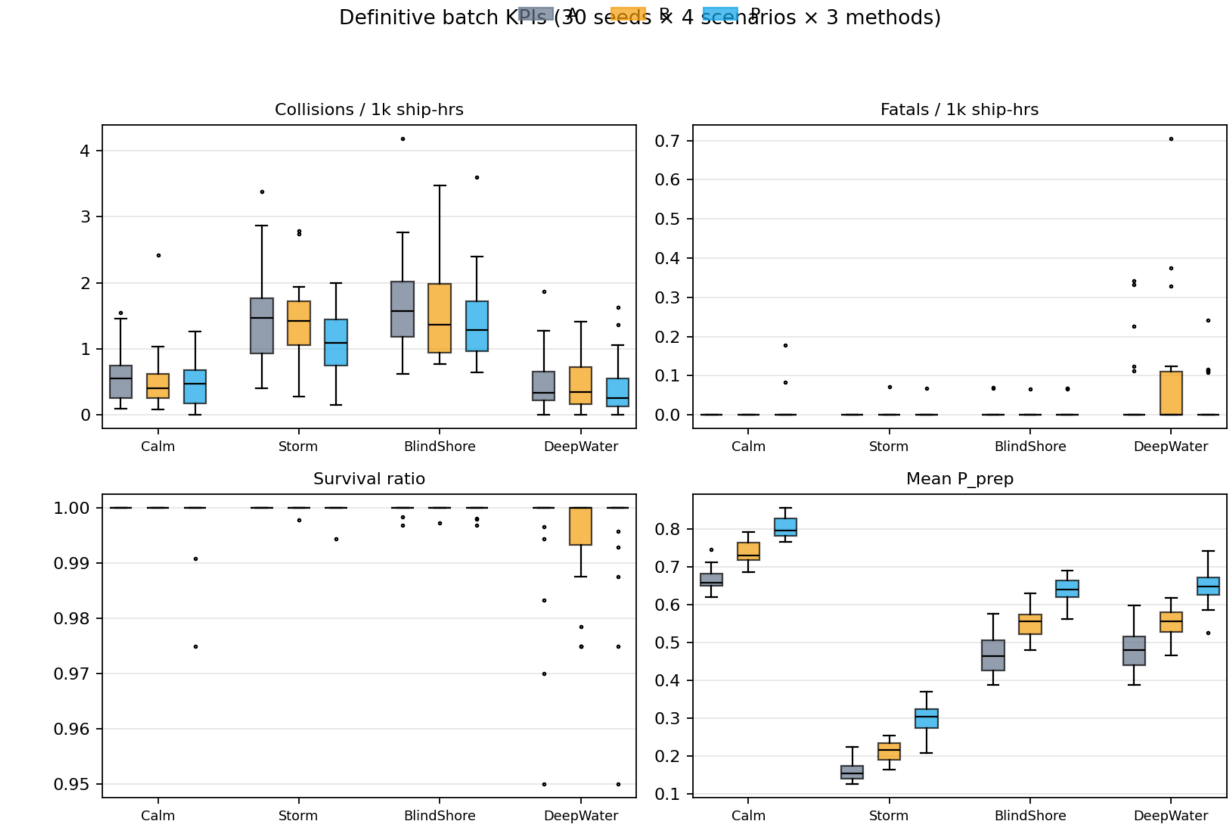

Figure 6: *Per-run KPI distributions for the definitive batch (30 seeds). A = Baseline A, B = Baseline B, P = Proposed.*

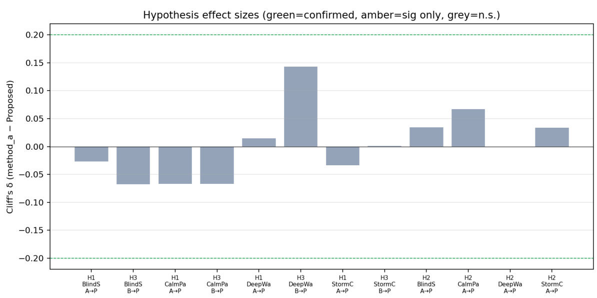

Figure 7: *Cliff's δ for H1–H3-relevant pairs. Green: confirmed; amber: significant only; grey: not significant after Holm.*

[12] P. A. M. Silveira, A. P. Teixeira, and C. G. Soares, "Use of AIS data to characterise marine traffic patterns and ship collision risk off the coast of portugal," *Journal of Navigation*, vol. 66, no. 6, pp. 879–898, 2013.

[13] W. Zhang, F. Goerlandt, P. Kujala, and Y. Wang, "An advanced method for detecting possible near miss ship collisions from AIS data," *Ocean Engineering*, vol. 107, pp. 60–69, 2015.

[14] C. Chauvin, S. Lardjane, G. Morel, J.-P. Clostermann, and B. Langard, "Human and organisational factors in maritime accidents: Analysis of collisions at sea using the HFACS," *Accident Analysis & Prevention*, vol. 59, pp. 26–37, 2013.

[15] A. Antelmi, G. Cordasco, D. D'Ambrosio, D. De Vinco, and C. Spagnuolo, "Reliable and efficient agent-based modeling with krABMaga," *Journal of Artificial Societies and Social Simulation*, vol. 27, no. 2, p. 4, 2024.

[16] EMODnet Bathymetry Consortium, "EMODnet digital bathymetry (DTM 2024)," European Marine Observation and Data Network (EMODnet), Bathymetry portal, 2024, gridded water-depth product at $1/16'$ ($\approx$115 m) resolution; retrieved via the OGC WCS endpoint `ows.emodnet-bathymetry.eu/wcs`. [Online]. Available: https://emodnet.ec.europa.eu/en/bathymetry

[17] EMODnet Human Activities, "Vessel density map," European Marine Observation and Data Network (EMODnet), Human Activities portal, 2023, AIS-derived ship density (hours km^{-2} month^{-1}) on a 1 km grid, resolved by vessel type; retrieved via the OGC WCS endpoint `ows.emodnet-humanactivities.eu/wcs`. [Online]. Available: https://emodnet.ec.europa.eu/en/human-activities

[18] PIANC, "Harbour approach channels design guidelines (MarCom Report no. 121)," World Association for Waterborne Transport Infrastructure (PIANC), Brussels, Belgium, Tech. Rep., 2014.

[19] P. E. Hart, N. J. Nilsson, and B. Raphael, "A formal basis for the heuristic determination of minimum cost paths," *IEEE Transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, 1968.

[20] F. Xiao, H. Ligteringen, C. van Gulijk, and B. Ale, "Nautical traffic simulation with multi-agent system," in *Proceedings of the 16th International IEEE Annual Conference on Intelligent Transportation Systems (ITSC)*. The Hague, The Netherlands: IEEE, 2013, pp. 1245–1252.

[21] P. T. Pedersen and S. Zhang, "On impact mechanics in ship collisions," *Marine Structures*, vol. 11, no. 10, pp. 429–449, 1998.

Table 5: *Selected engine default constants.*

Constant	Default	Role
Δt	0.25 h	tick length (15 min)
T	2880	ticks per run (30 d)
weather grid	50×50	hazard field
(w_s, w_v, w_u)	$(1.0, 0.5, 0.8)$	hazard fusion weights
warn / CPA / TTA gate	20 / 5 nm / 12 tk	avoidance gate
L_f, L_o	3 / 5 nm	give-way waypoint offsets
cooldown	3 tk	re-manoeuvre lockout
r_{coll}	0.05 nm	hit at $2r = 0.10$ nm
port collision exclusion	1 nm	harbour water
follow vicinity / gap	3 / 0.5 nm	speed-match / keep-distance
same-destination	5 nm	endpoint match radius
$(m_{\text{sway}}, m_{\text{surge}})$	$0.85, 0.05$	SIMCOL added mass
Minorsky a, b	$47.2, 32.7$	energy–volume (MJ, m^3)
$\ell/L_{pp}, \tau$	0.12, 0.08 m	damage length / side steel
$r_0, r_1, S^\dagger$	$0.10, 0.30, 0.50$	S_i knee / loss / founder
$(\beta_{\text{base}}, \beta_W, \beta_c)$	$0.0015, 0.0040, 0.0008$	fatigue rates
recovery	0.025/tk	dock fatigue recovery
F^*, κ	0.40, 0.65	comms-degradation knee
storm-comms (corridor)	0.08	Storm Corridor link rate
storm-comms (generic default)	0.30	when storm enabled outside corridor presets
n crew	10	per vessel
dwell	8–48 tk	port stay
helicopter / patrol	80 / 25 kn	rescue asset speeds
Δt_{mob}	2 tk	rescue mobilisation delay
helo / patrol cutoff	40 nm	SAR asset-selection range
w, r_0, u_d	5 nm, 2 nm, 0.2 nm/tk	MASSIM sweep / datum / drift
survival window	96 tk	liferaft endurance (24 h)
τ_{mob}	12 tk	MOB cold-water window (3 h)
MOB scatter	0.5 nm	person-in-water spread
P_c (IWRAP)	0.5–1.3$\times 10^{-4}$	causation probability

[22] V. U. Minorsky, "An analysis of ship collisions with reference to protection of nuclear power plants," *Journal of Ship Research*, vol. 3, no. 2, pp. 1–4, 1959.

[23] International Maritime Organization, "SOLAS chapter II-1: Construction — subdivision and stability (probabilistic damage stability)," International Maritime Organization, London, UK, International Convention for the Safety of Life at Sea, 2020.

[24] Z. Lusic and M. Ćorić, "Models for estimating the potential number of ship collisions," *Journal of Navigation*, vol. 68, no. 4, pp. 735–749, 2015.

[25] P. T. Pedersen, "Collision and grounding mechanics," *Proceedings of the Danish Society of Naval Architecture and Marine Engineering (WEMT)*, pp. 125–157, 1995.

[26] N. Cliff, "Dominance statistics: Ordinal analyses to answer ordinal questions," *Psychological Bulletin*, vol. 114, no. 3, pp. 494–509, 1993.

A. Default Constants

Table 5 lists principal defaults (overridable in configuration).

Weather-field process (Mixed preset): regime flips $p_{cs} = 0.03$, $p_{sc} = 0.08$; Calm/Storm reversion targets $(0.30, 0.80, 0.25)$ / $(0.72, 0.30, 0.75)$; up to 8 storm cells; AR(1) coefficients $(\rho_s, \rho_v, \rho_u) = (0.95, 0.90, 0.93)$, $(\sigma_s, \sigma_v, \sigma_u) = (0.07, 0.06, 0.07)$; gradient cap 0.12. Calm and Stormy presets scale the flip rates.